Kandala Ajit Prasad

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.84
FIITJEE JUNIOR COLLEGE	Class XII (BOARD OF INTERMEDIATE EDUCATION ANDHRA PRADESH)	2018	94.50%
NAVODAYA HIGH SCHOOL	Class X (BOARD OF SECONDARY EDUCATION ANDHRA PRADESH)	2016	9.20%

WORK EXPERIENCE

Project Assistant | Indian Institute of Science (IISc), Bengaluru

- Fabricated a **1-DOF Scheimpflug adapter** for INR 70,000, a **65% cost reduction** versus the INR 2 lakh commercial equivalent. Designed a calibration plate for INR 15,000, an **81% cost reduction** versus the INR 80,000 commercial equivalent.
- Developed a **6 m long flow-through pipe experimental setup** integrating Stereo PIV, high-speed imaging, and Scheimpflug optics, achieving stable laminar flow at Re 7,000 through controlled disturbance suppression.
- Designed CAD models for the PIV box, pipe couplers, and calibration plate, and performed **Stereo PIV** alignment, optical calibration, and post-processing for flow-field measurements.

PROJECTS

Computational and Analytical Investigation of Electrospinning Jet Instabilities | M.Tech Thesis

- Spearheaded a coupled computational and analytical investigation into electrospinning liquid jet instabilities under varying electric fields, formulating a **Linear Stability Analysis** (LSA) model from first principles to theoretically predict instability metrics.
- Executed **OpenFOAM-based CFD simulations** to model jet behavior across diverse operating conditions, including voltage, domain length, charge density, and permittivity.
- Engineered a **3D non-axisymmetric interface-tracking** pipeline in MATLAB to bridge numerical and theoretical data, developing centerline-extraction algorithms to compute CFD-derived growth rates for direct quantitative validation against LSA predictions.

Rayleigh-Plateau Instability | Mini Project

- Simulated a multiphase **Rayleigh-Plateau breakup** of a charged Newtonian jet in OpenFOAM using VOF method.
- Wrote **custom MATLAB code** to track the jet interface and compute perturbation growth rate from the raw simulation data, achieving an exact match between simulated and theoretical growth rates .
- Benchmarked the computed growth rate against published data Newtonian jets (mode m = 0), validating the analysis pipeline as the foundation for the subsequent M.Tech electrospinning project.

Self-Balancing Bicycle Using Reaction Wheel | B.Tech Project

- Simulated a reaction-wheel-balanced bicycle as an inverted pendulum in MATLAB, investigated the effect of varying the center-of-gravity height on destabilizing torque and stability requirements.
- Stabilized the bicycle from a disturbance angle of 23° using a **state space feedback controller**, achieving upright in 2 s, settled within 7 s through MATLAB simulations**.**
- **Designed** the bicycle and reaction-wheel assembly in Fusion and **fabricated** a balsa-wood prototype to demonstrate the mechanical implementation and feasibility of reaction-wheel stabilization.

INTERNSHIPS

Scientific Intern | Indian Institute of Science (IISc), Bengaluru

- Validated Basler cameras (up to 200 fps) for **PIV** flow imaging using an **Arduino-driven LED circuit** (no dropped frames); configured a **LabVIEW** VI to automate 1 Hz data acquisition from a programmable multimeter.
- Simulated the transition to chaos in Lorentz-force-driven 2D flow in MATLAB, mapping **periodic, quasi-periodic, and chaotic regimes** with state-space plots and corroborating them against experimental velocity data.
- Built a thermistor logger (30 s intervals, 12 hours) and an **LM317 constant-voltage supply**; helped set up 532 nm Nd:YAG laser-sheet optics (Powell lens) and PIV with polyamide seeding particles.

R&D Intern, Air-Breathing Rotating Detonation Engine (RDE) | Prime Toolings

- Simulated steady-state CFD on a commercial software (**ANSYS Fluent**) on an annular flame holder at 21 m/s, resolving recirculation zones and wake structure for flame stabilization.
- Performing **k-epsilon CFD** of a missile body and fin at Mach 2.25 and 17° AoA, mapping pressure loads to Al 6061 FEA, and FEA of a 220 N thrust test rig and a 12 bar SS 316 chamber.
- Preparing for RDE research on **thermo-acoustic** instability and supersonic **fuel-air mixing** and combustion chamber casing **vibration**.

RELEVANT COURSES

Thermal and Fluids: Computational Fluid Dynamics, Advanced Heat Transfer, RAC, ICE, Incompressible Fluid Flow, Interfacial fluid flows.

Numerical and Data-Driven: Machine Learning, Deep Learning, Perturbation Methods, Linear Stability Analysis, Non-Linear Dynamics.

SKILLS

Technical Skills: Fusion, Ansys Workbench, Ansys Fluent, ICEM CFD , OpenFOAM, ParaFOAM, Python, MATLAB, Overleaf, MS Office.

Soft Skills: Teamwork, Communication, Problem Solving, Analytical Skills, Persistence, Attention to Detail.

POSITIONS OF RESPONSIBILITY

Teaching Assistant | ME1060 | ME5310 | IIT Hyderabad

- Assisted the course instructor in coordinating academic activities for a class of 70 students.
- Supported course administration and evaluation-related activities throughout the semester.

EXTRACURRICULAR

Recap Journal Club -- IIT Hyderabad

- Participating in regular research-paper discussions and literature reviews through the departmental journal club, with a focus on understanding recent developments in engineering research.

MILAN 2026, IIT Hyderabad - Represented Bhabha Hostel in Cricket**.**

HOBBIES

Cricket, Chess, Creative Writing, Fantasy and Science Fiction Reading, Film Appreciation.